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Detection and Classification of Respiratory Syndromes in Original and modified DCGAN Augmented Neonatal Infrared Datasets

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Abstract

In the current pandemic scenarios, a non-invasive method for determining a neonate's respiratory rate and categorizing them using a deep learning technique is highly pertinent. Acquiring an infrared neonatal dataset for detecting and classifying respiratory syndromes is challenging. The limited number of infrared videos and images representing different types of syndromes is a tremendous challenge to the accuracy of the deep learning model. This paper uses the Deep Convolutional Generative Adversarial Networks(DCGAN) with gradient penalty for the data augmentation. The Discriminator in a standard DCGAN architecture is a convolutional neural network (CNN) that receives an image as input and outputs a single scalar value that indicates the likelihood that the input image is real or fake. Adding a gradient penalty adds a regularisation term to the loss function. This modification helps to stabilize training by preventing mode collapse and generating higher-quality images. The augmented dataset helped to make the original imbalanced dataset more balanced and increased the size of the original dataset. When the accuracies of the deep learning models trained on the original and balanced augmented neonatal datasets were compared in this work, the model based on the balanced augmented dataset performed better.

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Keywords: Neonatal; Infrared imaging; DCGAN; Deep Learning; Respiratory rate

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1. Introduction

COVID-19 being one of the worst pandemics faced by humankind, every innovation after the pandemic thinks of having a non-invasive approach. A non-invasive approach will help humans monitor, detect, and analyze parameters without direct contact with the subject. Especially with infants, the non-invasive approach proves efficient for multiple reasons. Among them are infection-free monitoring, the comfortableness of the infants due to a less wired approach, and the ease of acquiring data.

The thermal cameras can monitor the neonates and effectively determine the respiratory rates to detect whether they have any respiratory syndromes[1] like Bradypnea, where the respiratory rate is less than normal, and Tachypnea, where the respiratory rate is more significant than normal. A deep learning model was developed [2] to effectively classify the respiratory rates and provide corrective feedback to the doctors for the immediate attention of the neonates. The major limitation of the deep learning model was the lack of more extensive and balanced datasets. The effort to make a bigger balanced dataset for the deep learning model was achieved by augmenting the existing dataset using a modified DCGAN with gradient penalty[3]. The neonatal dataset generated by the DCGAN model is then given to the deep learning model to detect and classify the respiratory syndromes. The results were compared with those obtained from the deep learning model using the original imbalanced dataset. Fig. 1 shows the system architecture.

The remaining paper is divided into three modules; the first module discusses the existing works and the concepts that contributed to the development of the model. The second module examines the DCGAN model that helped augment the infant dataset and the deep learning model that classifies the respiratory rate into different respiratory syndromes. The third module compares the results of the models trained using original and augmented datasets.

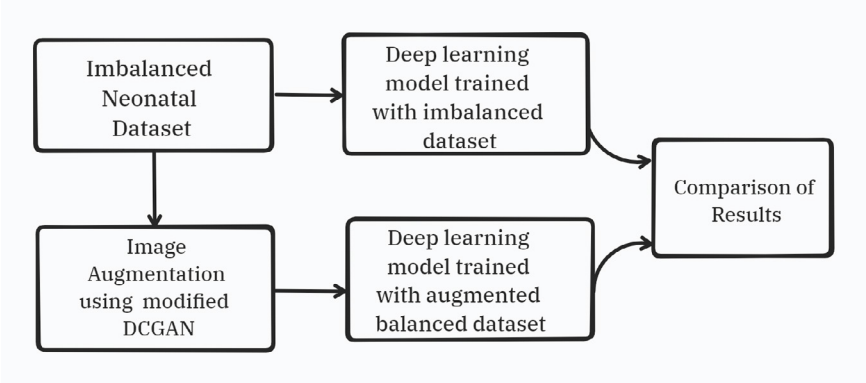


Fig. 1: System Architecture

2. LITERATURE REVIEW

Carina Barbosa et al.[1] suggested a black box approach for finding the respiratory rates of neonates using thermal images. Abbas K Abbas et al.[4] considered the nose region the initial Region Of Interest(ROI) and utilized a noncontact approach to monitor the respiratory rates using infrared images. The black box approach[1] was found to have better results than the single point of ROI [4] because in the black box approach, the ROI is not a defined region [1], instead, we consider the entire frame itself as a grid structure and each grid is considered to be a ROI[16]. Carina Barbosa Pereira et al.[5] used the nose region as ROI to detect and monitor breathing dynamics using thermography. C. Bruser et al. used a technique of speech processing to estimate the heart rate in ballistocardiograms[6].

Artur Grudzień et al. suggested a face verification method in thermal images using Convolutional Neural Network(CNN) [7]. Features extracted using CNN will be used to identify faces by comparing the distance metrics of the samples; the paper [7] used the Siamese CNN method to compare the capability with the existing CNN architectures. Fatih Ertam et al.[8] proved that the ReLU activation function gives the best classification rate after comparing different CNN and softMax classifiers. Jing Bin et al. created a model for detecting Neonatal Respiratory Distress Syndrome(NRDS)[9], giving an accuracy of 98.07%

The Convolutional Neural Networks[19] is a deep learning network that works well for image classification. We need an enormous, balanced dataset to get a good classification accuracy from a CNN model. The traditional data augmentation methods generated more images of the same class. The rotating, mirroring, zooming, and blurring will create more images. Ian J. Goodfellow et al. proposed Generative Adversarial Networks(GAN)[13] where two deep learning models were trained simultaneously. The generative model captures the data spread, and the discriminative model gives the probability of a sample, i.e., whether that sample is a part of training data or a generated one. This GAN[17] model proved to be useful for data augmentation.

Maayan Frid-Adar et al. [11] proposed a flawless liver lesion identification method where GAN-based data is augmented on liver lesions. The classification accuracy was 57% when no augmentation was applied, and accuracy increased to 78.6% when traditional augmentation techniques were used. The synthesized augmentation increased the classification[18] accuracy to 86.56%. When the radiologists checked the generated data, they could identify the real and fake lesions. One identified the generated and the real lesions with 77.5% and 78% accuracy, and the other classified the generated and real lesions with 69.2% and 69.2% accuracy, respectively. This proved that both the generated results were worthwhile. The paper helped establish that the GAN-augmented results are more accurate than traditional augmentation.

Qiufeng Wu et al. [12] uses a method that refines the detection rate of tomato leaf diseases by augmenting the dataset using GAN models. By employing a combination of DCGAN augmented images and the original images as inputs to the GoogLeNet model, a notable increase in detection accuracy was seen, reaching 94.33% as reported in the study[3]. The application of hyperparameter tuning resulted in enhancements to the convolutional neural networks. At the same time, a variety of Generative Adversarial Networks (GANs) were chosen to train and evaluate five distinct categories of tomato leaf images. Utilizing the DCGAN augmentation technique resulted in generating a dataset that exhibited greater comprehensiveness and equilibrium.

3. DATA AUGMENTATION AND CLASSIFICATION

The section is divided into three subsections; the first discusses the data augmentation part of the neonatal dataset, the second one discusses the extraction of respiratory signals from thermal images, and the third discusses the detection and classification of respiratory syndromes using a deep learning model.

3.1. Dataset Augmentation

The initial dataset[2] consists of two-minute infrared videos of infant. The infant is monitored in a closed premise for a passive thermography approach. The passive thermography approach is used because the backdrop will discharge less thermal radiation than any neonate kept under observation, giving precise readings from neonates. The infant's thermal images are obtained from their thermal videos captured under a restricted environment[2].

The count of unwell neonates in the dataset is significantly less when compared to that of good ones. In many cases, the cause of the illness is unknown or not monitored. The dataset comprises 8,000 infrared images of healthy neonates and 3,000 infrared images of unhealthy neonates at the outset. The imbalanced dataset will cause poor classification by a deep learning model. The classification accuracy will be very high even if an unwell neonate is classified as healthy because the count of wrongly classified images is significantly less.

At the same time, a balanced neonatal dataset would increase the model's accuracy because the classifier understands the features of unhealthy neonates, avoiding the wrong classification and overfitting. A DCGAN model can supplement the outnumbered group of unhealthy neonatal images to get a wide variety in the resultant dataset. Traditional augmentation techniques will create 18000 images, of which 12000 were of healthy infant class[10]. The following procedure will help to augment the dataset using DCGAN.

3.1.1. Pre-processing Stage

The dataset contains images of different resolutions. Each image has to be resized to 128x128 to be given as input to DCGAN. The lower the image resolution, the lesser the training time of the DCGAN. All the 18000 images are resized to 128x128. Figure 2 shows that the saple input of neontal thermal images.

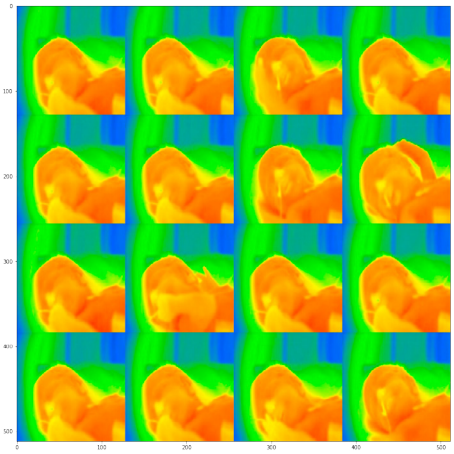


Fig. 2: A sample input of neonatal thermal images

3.1.2. *Training of DCGAN model for Data Augmentation*

The DCGAN model has a discriminator and generator neural network. The Discriminator in the DCGAN model is a CNN classifier that gives exact labels to a set of original and generated images. The generator neural network distinguishes the original images from those generated to produce original-like images. The model provides good accuracy when the Discriminator detects no difference between the original and created ones.

3.1.3. *Training the Discriminator Neural Network*

The discriminator neural network is a simple classifier network that takes images of the resolution 128x128 as input and has five convolutional layers, each with different filters which is shown in figure 3.

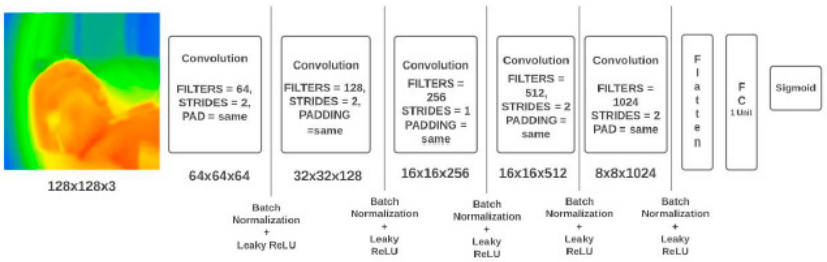


Fig. 3: Discriminator Network

In a discriminator neural network, each convolution layer of the network is followed by a Batch Normalization and a Leaky ReLU. The filter size in each convolutional layer increases from 64 to 128 to 256 to 512 to 1024, with a kernel size of 5x5. The output of the last convolution layer is fed to a dense layer, and the discriminator loss is calculated with a gradient penalty.

Gradient penalty is a regularisation approach that is often used to stabilize and improve the training of discriminator networks in deep convolutional generative adversarial networks (DCGANs) and other GAN variants. It's beneficial for avoiding mode collapse and ensuring that the discriminator gradients to the generator during training. When the generator produces just a small set of samples, mode collapse occurs, making the Discriminator's task too easy. The Discriminator produces a binary judgment for a given image, either 0 or 1. 0 is given for the fake image, and one is for the original image. The back propagation algorithm helps update the weights based on the loss calculated by the algorithm 1. By identifying the generated images as fake, the Discriminator provides feedback to the generator. This will push the generator to generate images similar to the originals. The Discriminator attempts to minimize the stochastic gradient, as shown in equation 1 for every given image.

$$logD(x) + log(1 - D(G(z))) \quad (1)$$

where x denotes the original thermal image of a newborn and G(z) denotes the generated thermal image of a neonate.

Algorithm 1 Training the Discriminator Neural Network

Input: Batch of real and generated neonatal thermal images
Output: Trained discriminator neural network

1. To assign real images with the prediction label based on the discriminator output.
2. To assign generated images with the prediction label based on the discriminator output.
3. Calculating discriminator loss with gradient penalty on real images.
4. Updating the discriminator weights on account of the loss calculated.
5. Calculating discriminator loss with gradient penalty on generated images.
6. Updating the discriminator weights on account of the loss calculated.
7. The label should be assigned 1 if the Discriminator successfully classifies the image as genuine. Otherwise, the label should be assigned 0.

3.1.4. Training the Generator Neural Network

The generator neural network can comprehend data distribution in real images, enabling it to generate fake images that closely resemble genuine ones. The organization of the generator neural network is depicted in Figure 4. The generator network can gain the training it requires to develop an understanding of mapping the latent vector space to generate images that closely resemble genuine ones (Reference 13). The neural network utilizes a noise vector (z) of 100 dimensions randomly sampled from a continuous or normal distribution of data points. The 65,536 units of the dense layer undergo a resizing operation to form an 8 x 8 x 1024 structure, which is then passed through a Leaky ReLU layer shown in figure 4.

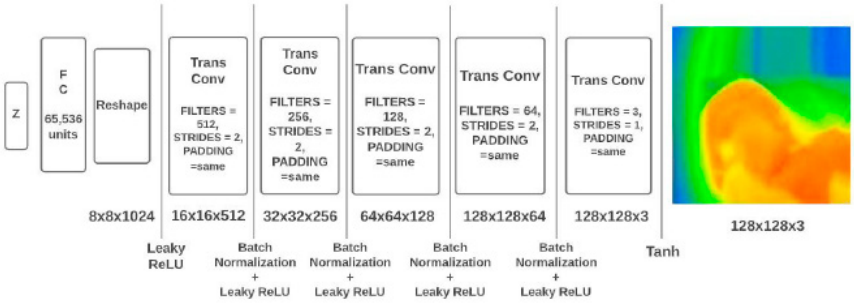


Fig. 4: Generator Network

Following each convolution layer, batch normalization and the Leaky ReLU layer are applied sequentially. The last transposed convolutional layer is equipped with three filters and operates with a stride value of one. The TanH activation layer is applied after the convolution layer. The discriminator neural network is fed with a 128x128 image generated and blended with the original thermal images. Subsequently, the weights are updated using the Backpropagation process. The generator network aims to minimize the loss for each generated image, as specified in equation 2[15].

$$log(1 - D(G(z))) \tag{2}$$

The loss of the generator neural network is calculated per the Discriminator's response, as shown in the algorithm 2.

Algorithm 2 Training Generator Neural Network

Input: A Set of latent vectors generated from random noise vectors
Output: Trained generator neural network

1. Samples taken from latent space are used to generate images.
2. According to the feedback from the discriminator network, prediction labels are assigned to generated data.
3. The generator loss is computed by summing the logarithmic differences between 1 and the expected label.
4. The generator weights have been updated to account for the calculated loss.

3.1.5. Augmented Dataset

A value of 0.002 is set as the learning rate for the Discriminator, and a value of 0.0002 is set as the learning rate for the generator throughout the training process. The lower learning rates ensure the Discriminator is not very competitive in learning. When the Discriminator becomes competitive, the generator will fail to produce good images because the Discriminator will decline the generated images with high trust values. A 0.0002 is given as the learning rate of the generator. The DCGAN model was executed for 500 epochs, each with a batch size 32 which is shown in figure 5.

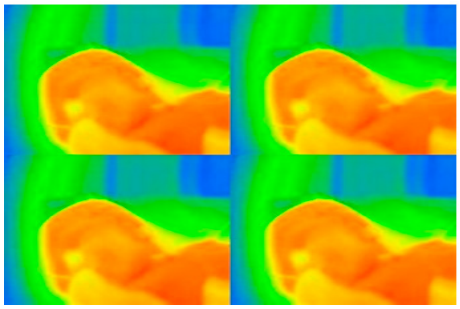


Fig. 5: Sample Output from epoch 225 and batch size of 32

Other parameters, such as the Adam optimizer's learning rate and momentum, are set to 0.2 and 0.5, respectively. The discriminator network generated the best loss of 0.7055, while the generator network generated the best loss of 0.6932. A binary cross-entropy [1] loss function uses a gradient penalty for calculating discriminator loss. A loss value of about 0.6 suggests that the Discriminator cannot distinguish between actual and fraudulent photos. This means that the generator was capable of producing excellent original-like images. The data spread for these generated images is the same as for the original images. We could produce a larger balanced dataset with our DCGAN model. More than 15000 healthy and 15000 unhealthy neonatal thermal images were included in the generated neonatal thermal dataset.

3.2. Extracting Respiratory Signals from Thermal Images

The black box approach[1] is used to extract the respiratory signals from a neonatal image. The ROI is not a single point of interest; the whole image is divided into a group of 100 grids, and each grid is considered a point of interest or ROI. The signals whose Signal Quality Index(SQI) is above the threshold value of 0.5 are selected as a respiratory signal, and the respiratory rate is calculated as given in the algorithm 3[2].

3.3. *Deep Learning model for Respiratory Syndrome Classification*

Now that we have a bigger balanced dataset, the next task is to use this dataset to create and train a deep learning model and classify the respiratory syndromes. The augmented and original datasets are used to train two different deep learning models, and later, the results are compared to see which dataset gives more accuracy.

Algorithm 3 Extraction of respiratory rate from Thermal Images

Input: Neonatal thermal images
Output: Respiratory rate from signals

1. Apply a grid onto the input image, which will then split the input image into 100 Regions Of Interest.
2. The mean pixel value of every Region Of Interest is calculated to construct a respiratory signal.
3. Use a hamming window for mean centering and convert it to the frequency domain.
4. Using the Low Pass, Band Pass, and High Pass filters, calculate the SQI of signals.
5. For signals with SQI above 0.5, the maximum value is considered and multiplied by 60 to calculate the respiratory rate,
6. The respiratory rates are then saved.

The sequential deep learning model comprises two hidden layers, a ReLU layer and a softmax layer. The model classifies the detected respiratory rates as Healthy, Bradypnea[1], Tachypnea[1] and No Information. The model is trained for 500 epochs. The algorithm 4 explains how the classification is done with a deep learning model.

Algorithm 4 Deep Learning model for respiratory syndrome classification

Input: Respiratory Signals
Output: Classes of Respiratory Syndromes

1. The dataset is partitioned into an 80:20 ratio.
2. A first layer of 225 nodes is created using Keras.
3. Two 100-node hidden layers are created, followed by the activation function.
4. Class labels get assigned to the identified respiratory signals.
5. The class labels are subjected to one-hot encoding and passed to the softmax layer.
6. One 4-node output layer with a softmax activation function is created using Keras.
7. The model is compiled using the Adam optimizer and employing categorical cross-entropy as the loss calculation method.

4. Results and Analysis

Every healthy infant has a respiratory rate between 40 and 60 breaths per minute. If the rate is above 60 breaths per minute, the infant has Tachypnea[1], and if the rate is below 40 breaths per minute, the infant has Bradypnea[1] syndrome. The deep learning model trained with the original dataset and another one of the same architecture trained with the DCGAN augmented dataset will broadly classify these respiratory rates. The confusion matrix for the deep learning model using an augmented dataset is shown in figure 6b. The model has an accuracy of 93.45% and an

F1 score of 93.41%, according to the confusion matrix. The recall is 95.46% and the precision is 91.59% for the healthy class. Likewise, Bradypnea, a respiratory condition, is classified with 98.10% precision and 95.19% recall. The precision and recall of the no-information class were 79.74% and 72.79%, respectively.

Confusion Matrix

TARGET / OUTPUT	Healthy	Bradypnea	No Information	SUM
Healthy	10471 58.17%	982 5.46%	547 3.04%	12000 87.26% 12.74%
Bradypnea	41 0.23%	4815 26.75%	250 1.39%	5106 94.30% 5.70%
No Information	4 0.02%	41 0.23%	849 4.72%	894 94.97% 5.03%
SUM	10516 99.57% 0.43%	5838 82.48% 17.52%	1646 51.58% 48.42%	16135 / 18000 89.64% 10.36%

(a) Confusion matrix(Original Dataset)

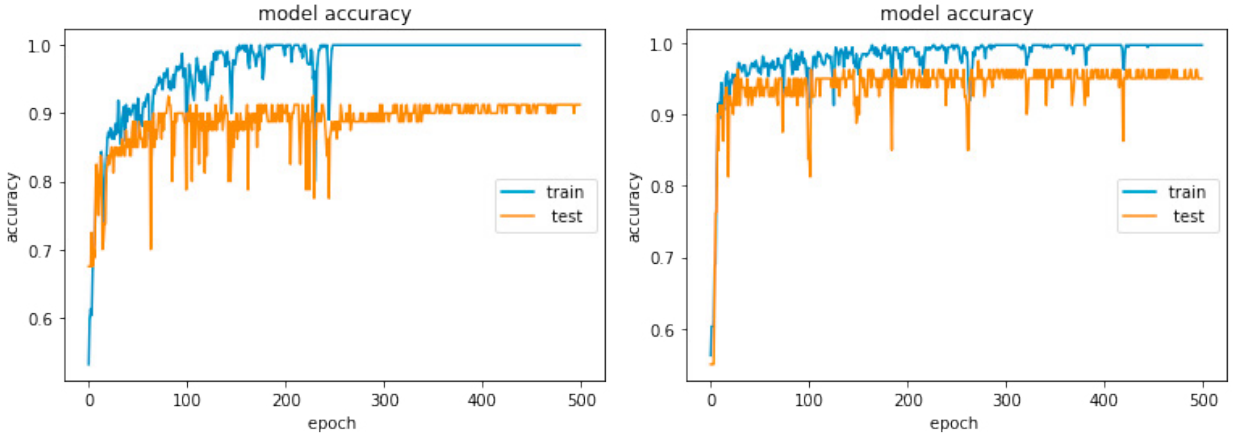
Confusion Matrix

TARGET / OUTPUT	Healthy	Bradypnea	No Information	SUM
Healthy	13739 45.80%	580 1.93%	681 2.27%	15000 91.59% 8.41%
Bradypnea	241 0.80%	12474 41.58%	0 0.00%	12715 98.10% 1.90%
No Information	412 1.37%	51 0.17%	1822 6.07%	2285 79.74% 20.26%
SUM	14392 95.46% 4.54%	13105 95.19% 4.81%	2503 72.79% 27.21%	28035 / 30000 93.45% 6.55%

(b) Confusion matrix(Augmented Dataset)

Fig. 6: Comparison of Confusion matrix

Figure 6a shows the confusion matrix of the deep learning model trained using the Original dataset. According to the confusion matrix, the model has an accuracy of 89.64% and an F1 score of 88.9%. For the healthy class, the precision is 87.26%, and the recall is 99.57%. Similarly, the respiratory disease bradypnea is categorized with 94.30% precision and 82.48% recall. The no-information class had a precision of 94.97% and a recall of 51.58% , respectively.

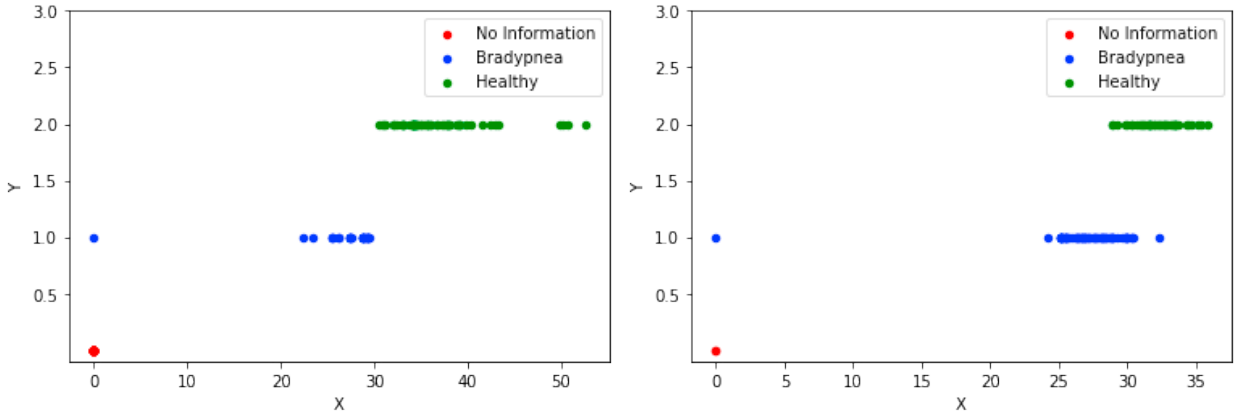


(a) Model Accuracy(Original Dataset)

(b) Model Accuracy(Augmented Dataset)

Fig. 7: Comparison of Model Accuracy

The model accuracy is shown in figures 7a and 7b. The scatter plot for the models trained with the original and augmented dataset is shown in figure 8a and 8b. The respiratory rates will be on the X-axis, and the Y-axis represents the classes. The value 0 on the Y axis represents ”No information”, 1 represents ”Bradypnea”, 2 represents ”Healthy,” and 3 represents ”Tachypnea”. Since the dataset does not have Tachypnea samples, the class is not in the scatter plot.



(a) Scatter Plot (Original Dataset)

(b) Scatter Plot (Augmented Dataset)

Fig. 8: Comparison of Scatter Plots

A sample snapshot of the respiratory signals generated using the original and augmented dataset, its SQI index, the respiratory rate calculated, and the respiratory syndrome identified are shown in table 1.

Original Dataset				**Augmented Dataset**			
Signal	SQI	Respiratory Rate	Respiratory Syndrome	Signal	SQI	Respiratory Rate	Respiratory Syndrome
Sig1	0.57	50.93	Healthy	ASig1	0.67	32.93	Bradypnea
Sig2	0.75	30.42	Bradypnea	ASig2	0.81	34.42	Bradypnea
Sig3	0.65	48.93	Healthy	ASig3	0.75	58.93	Healthy
Sig4	0.17	0	No Information	ASig4	0.54	32.55	Bradypnea
Sig5	0.27	0	No Information	ASig5	0.57	44.11	Healthy
Sig6	0.51	55.93	Healthy	ASig6	0.16	0	No Information
Sig7	0.72	25.93	Bradypnea	ASig7	0.75	35.93	Bradypnea
Sig8	0.57	20.22	Bradypnea	ASig8	0.15	0	No Information
Sig9	0.8	50.98	Healthy	ASig9	0.63	55.98	Healthy
Sig10	0.42	0	No Information	ASig10	0.52	30.34	Bradypnea

Table 1: Sample Output

5. Conclusion

In this paper, a comparative study is done on the performance of the deep learning models trained using the imbalanced original neonatal dataset and the one trained using the modified DCGAN augmented balanced dataset. Both models are trained for the same number of epochs and with the same learning rate. The model trained with an augmented dataset with a balanced ratio of classes outperformed the model with the original dataset with the imbalanced dataset. The DCGAN model can be used to augment thermal datasets. It has been proved that the accuracy of deep learning models trained with such datasets can be better than those trained with the original dataset having imbalanced

classes. The augmented dataset can also be added to the original dataset, creating more extensive datasets for deep learning models. The same approach can be applied to other medical datasets, where the quality and the number of thermal images will be less.

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